

MTH 335: Introduction to Abstract Algebra Notes

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Chapter 1

Introduction to Sets

Set theory is a foundation on which (almost) all of mathematics can be built.

- A *set* is a collection of distinct objects called *elements*.

This is not really a definition (for a rigorous definition see this [link](#)); to avoid some messiness we will use this “intuitive” (sometimes called “naive”) understanding of set and element. Which has good enough for Cantor, so it should be good enough for us.¹

1.1 Describing Sets

Lots of new notation will be introduced as we move through the material. The trade-off for learning all the new notation is certainly worth it. Think of trying to solve even a simple quadratic, $x^2 + 2x - 10 = 0$, without the advantage of using x as the unknown and Arabic numbers.

- Sets are usually denoted with capital letters, e.g., $A, B, C, \dots$ etc; and elements in sets are usually denoted with lowercase letters $a, b, c, \dots$
- If a is an element in the set A , we write $a \in A$. If a is not in A we write $a \notin A$ (slashes usually mean negation).

Sets are described in a number of different ways:

Described:

Example 1.1. a, b, c are elements of the set A ; also written $a, b, c \in A$.

Example 1.2. $\mathbb{R}$ is the set of all real numbers.

Tabular form:

Example 1.3. $A = \{a, b, c\}$, means $a, b, c \in A$.

Dots are used when unlisted elements can be implied:

Example 1.4. $A = \{a, b, c, \dots z\}$, it is implied that all the letters in alphabet are in A .

Example 1.5. $\mathbb{Z} = \{\dots, -1, 0, 1, 2, \dots\}$

¹George Cantor founded set theory using the “naive” definitions in 1874 with a single paper. However by 1900 a number of contradictions were found. Most famously Russell’s paradox: “the set of all sets that are not members of themselves.” This set must be a member of itself, and yet is not a member of itself!

Set builder form:

Example 1.6. $\mathbb{Z}^+ = \{n \in \mathbb{Z} : n > 0\}$, read “ n is an integer *such that* n is greater than 0.”

Example 1.7. $2\mathbb{Z} = \{n \in \mathbb{Z} : n = 2m \text{ for some } m \in \mathbb{Z}\}$. This set is all the even numbers. So $4 \in 2\mathbb{Z}$ and $11 \notin 2\mathbb{Z}$.

1.2 Some Basic Definitions

A farmer had a 100 feet of fence and wanted to enclose as large an area as possible. After a little thought, an engineer made a square resulting in an area of $25^2 = 625 \text{ ft}^2$. A physicist (knowing he could do better) made a circle resulting in $\pi(100/2\pi)^2 \approx 795.7 \text{ ft}^2$. Finally a mathematician steps forward, breaks off a small piece of fence, and says “I define myself to be on the outside!” (resulting in an area approximately equal to the area of the Earth).

How we define things is very important. When trying to prove something, read the definitions very carefully. For the following definitions, assume that A and B are both sets.

Definition 1.1. A is a *subset* of B if every element of A is also an element of B , written $A \subseteq B$.

Example 1.8. If $A = \{1, 2, 3\}$ and $B = \{0, 1, 2, 3\}$ then $A \subseteq B$ and $B \not\subseteq A$. Also note that $A \subseteq A$.

- $A \subseteq A$ for any set A . Every set is subset of itself. Since every element of a set A will also be an element of A .

Example 1.9. $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$: The natural, integers, rational, real, and complex numbers respectively.

Definition 1.2. The set with no elements is called the *empty set*, denoted $\emptyset$.

- $\emptyset = \{x \in A : x \notin A \text{ for any set } A\}$.
- Let B be set of numbers that are both irrational and rational real numbers. $B = \emptyset$.
- $\emptyset \subseteq A$ for any set A , i.e., *the empty set is a subset of every set*.

Definition 1.3. $A = B$ if and only if A and B have the same elements.

- It is often helpful to use the equivalent definition: $A = B$ is $A \subseteq B$ and $B \subseteq A$.

Example 1.10. Let $A = \{a : a = 2^n \text{ for any integer } n\}$ and $B = \{b : b = 2^{n-2} \text{ for any integer } n\}$. Show that $A = B$. (Show that $A \subseteq B$) Let $a \in A$. If n is an integer then there is another integer m such that $m = n - 2$. So then $a = 2^n = 2^{m+2} \in B$.

(Show that $B \subseteq A$) Let $b \in B$. If n is an integer then there is another integer m such that $n - 2 = m$. So then $b = 2^{n-2} = 2^m \in A$.

- Note that proving $B \subseteq A$ was nearly identical to proving $A \subseteq B$ (I literally copied, pasted, and switch a few letters). When one part of prove is so clearly similar to an already completed previous part, we will often just write “similarly it can be shown that $B \subseteq A$ ”.

Definition 1.4. A is a *proper subset* of B if every element of A is also an element of B and $A \neq B$, written $A \subset B$.

Example 1.11. Let $A = \{a \in \mathbb{Z} : a = 2^n \text{ for a positive integer } n\}$ and $B = \{a \in \mathbb{Z} : a = 2n\}$. Show that A is a proper subset of B .

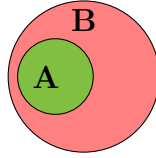


Figure 1.1:
 $A \subseteq B$ and $A \subset B$

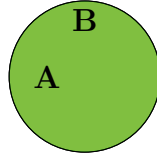


Figure 1.2:
 $A \subseteq B$ and $A \not\subset B$

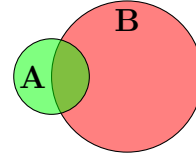


Figure 1.3:
 $A \not\subseteq B$

Figure 1.4: Venn diagrams

We must show that $A \subseteq B$ but $A \neq B$. Let $a \in A$. by definition $a = 2^n = 2 \cdot 2^{n-1}$ for $n \geq 0$. Since 2^{n-1} is an integer, $a \in A$. (To show that $A \neq B$ we need only find 1 element in B that is not in A . There are many.) $6 = 2 \cdot 3 \in B$ but $6 \notin A$ since $6 \neq 2^n$ for n a positive integer.

- Our use of $\subseteq$ and $\subset$ is analogous to $\leq$ and $<$.²

Definition 1.5. Let A be any set. The set of all subsets of A is called the *power set* of A , denoted $\mathcal{P}(A)$.

Exercise 1.1. Let $A = \{0, 1, 2, 3\}$. Find $\mathcal{P}(A)$ (don't forget A and $\emptyset$).

- Note that: $A \in \mathcal{P}(A)$, $\emptyset \in \mathcal{P}(A)$, and $\mathcal{P}(\emptyset) = \emptyset$.
- Theorem 1.1: If A be any set with n elements then $\mathcal{P}(A)$ has 2^n elements.

Remark 1.1. Some summary.

1. $A = B \iff A \subseteq B$ and $B \subseteq A$.³
2. $A \subseteq A$
3. $\emptyset \subseteq A$
4. $A \in \mathcal{P}(A)$ and $\emptyset \in \mathcal{P}(A)$

1.3 Set Operations

Definition 1.6. The *union* of sets A and B is defined by $A \cup B = \{x: x \in A \text{ or } x \in B\}$

Example 1.12. Let $A = \{1, 2, 3\}$ and $B = \{2, 3, 4\}$. $A \cup B = \{1, 2, 3, 4\}$.

Exercise 1.2. $A \cup B = B$ if and only if $A \subseteq B$.

Example 1.13. $\mathbb{R} \cup \mathbb{Z} = \mathbb{R}$. $\mathbb{Z} \subseteq \mathbb{R}$ so this is true by the above result.

Exercise 1.3. $A \cup B = B \cup A$.

Definition 1.7. The *intersection* of sets A and B is defined by $A \cap B = \{x: x \in A \text{ and } x \in B\}$.

Example 1.14. Let $A = \{1, 2, 3\}$ and $B = \{2, 3, 4\}$. $A \cap B = \{2, 3\}$.

Example 1.15. Let $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$. $A \cap B = \emptyset$.

Exercise 1.4. $A \cap B = B$ if and only if $B \subseteq A$.

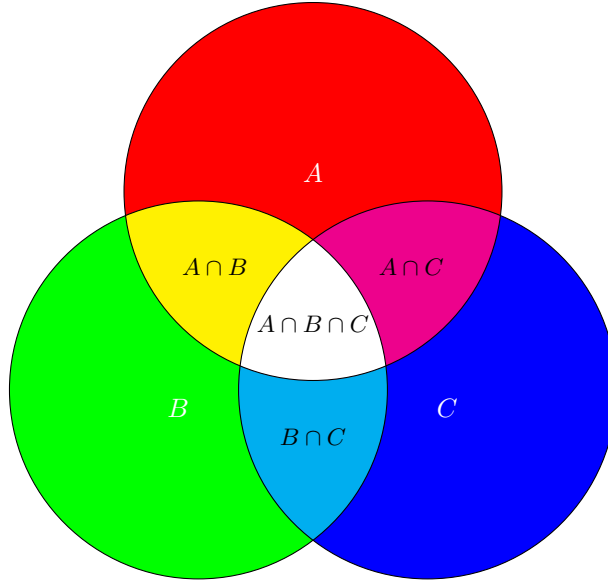


Figure 1.5: A Venn diagram showing the intersections of three sets, A , B , and C .

Exercise 1.5. $A \cap B = B \cap A$.

Definition 1.8. The *difference of sets* B and A is the set of elements that are in B and not in A ; written $B - A$.⁴

- $B - A = \{x: x \in B \text{ but } x \notin A\}$ (set builder form).

Example 1.16. $\mathbb{Z} - 2\mathbb{Z}$ is the set of odd integers.

Definition 1.9. The complement of a set A is the set of elements (under consideration) not in A ; denoted A^c .⁵

- $A^c = \{x: x \notin A\}$.

Example 1.17. When considering real numbers, $\mathbb{Z}^c = \{x: x \notin \mathbb{Z}\}$ is the set of all numbers that are not integers.

Proposition 1.1. $A - B = A \cap B^c$.

Proof. Let $x \in A - B$ so then $x \in A$ and $x \notin B$. By definition of compliments $x \in B^c$. So then $x \in A \cap B^c$ and $A - B \subseteq A \cap B^c$.

Let $x \in A \cap B^c$ so then $x \in A$ and $x \notin B$. By definition of difference of sets, $x \in A - B$ thus $A \cap B^c \subseteq A - B$.

$A - B \subseteq A \cap B^c$ and $A \cap B^c \subseteq A - B$ imply $A \cap B^c = A - B$. □

Definition 1.10. The number of elements in a set A is called the *order of* A ; denoted $|A|$.⁶

²Note that the meaning of set notation varies among authors. $\subsetneq$ is often used to denote proper subset, and others use $\subset$ for subset.

³“ $\iff$ ” is equivalent to writing “if and only if”. Often this is notated “iff.”

⁴Also called the *relative complement of* A in B usually denoted $B \setminus A$.

⁵We are being a little sloppy here by ignoring some complications of the universal set in trade for an easier introduction to sets.

⁶Also called the *cardinality of* A .

- When we write $|A| = n$ it is assumed that n is a non-negative integer.

Example 1.18. Let $A = \{a, b, c\}$. $|A| = 3$.

Example 1.19. Let $A = \{a, b, c\}$ then $|\mathcal{P}(A)| = 2^3 = 8$ by Theorem 1.2.

Example 1.20. $|\emptyset| = 0$.

Lemma 1.1. If $|A| = n$ and $|B| = n + 1$ for sets A and B then $|\mathcal{P}(B)| = 2|\mathcal{P}(A)|$.

Proof. We can without loss of generality⁷ assume that $B = A \cup \{b\}$. So that $A = B - \{b\}$. B has two kinds of subsets; those containing b and those that do not. Now let $S \in \mathcal{P}(B)$ such that $b \notin S$. Then $S \in \mathcal{P}(A)$, and there are n such sets. Now let $S' \in \mathcal{P}(B)$ such that $b \in S'$. Then $S' - \{b\} \subseteq A$, and there are n such sets. As a subsets of B either contains b or it does not, $|\mathcal{P}(B)| = |\mathcal{P}(A)| + |\mathcal{P}(A)| = 2|\mathcal{P}(A)|$. $\square$

Theorem 1.1. $|\mathcal{P}(A)| = 2^n$ for any set A .

Example 1.21. Prove theorem 1.2.

Hint: Use lemma 1.1 and induction.

1.4 Exercises

1. Write in tabular form:

- $A = \{x: x \text{ is a letter in "do not repeat"}\}$
 - $3\mathbb{Z} = \{x \in \mathbb{Z}: x = 3n \text{ for some } n \in \mathbb{Z}\}$
 - $N = \{x: x \text{ is odd and } x \text{ is even}\}$ (recall that 0 is even)

(b) Write these sets in set-builder form:

- Let $A = \{3, 4, 5, 6, 7, \dots\}$
- Let $B = \{3, 5, 7, 9, \dots\}$
- Let $\mathbb{Q}$ be the set of rational numbers.

2. Find all sets of $A = \{1, 2, 3\}$.

3. Prove that every nonempty subset has a proper subset.

4. Prove that if A is a subset of $\emptyset$, then $A = \emptyset$.

5. Prove that $A = \{2, 3, 4, 5\}$ is not a subset of $B = \{x \mid x \text{ is even}\}$.

6. Prove that A and B are subsets of $A \cup B$.

7. Prove that $A = A \cup A$ and $A = A \cap A$.

8. Prove that $A \cap B$ is a subset of A and of B .

9. Prove that $A \cup B = \emptyset$ implies that $A = \emptyset$ and $B = \emptyset$.

10. Prove that $A \cup B = B$ if and only if $A \subseteq B$.

11. Prove that $A \cap B = B$ if and only if $B \subseteq A$.

⁷Often abbreviated WLOG, "without loss of generality" is used before an assumption in a proof to narrow the premise to some special case when the proof using the special case can obviously be adapted for all other cases.

12. Prove that $A \cup (A \cap B) = A$.
13. Prove that $A \cap (A \cup B) = A$.
14. Prove that $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$.
15. Prove that $A - B$ is a subset of $A \cup B$.
16. Prove that if A is a subset of B then $A \cap B = A$.
17. Let $A \cap B = \emptyset$. Prove that $B \cap A' = B$.
18. Prove that if A is a subset of B then $A \cup B = B$.
19. Prove that $A' - B' = B - A$.
20. Prove that if A is a subset of B then B' is a subset of A' .
21. Prove that if $A \cap B = \emptyset$ then $A \cup B' = B'$.
22. Prove that $(A \cap B)' = A' \cup B'$.
23. Prove that $A \subset B$ implies that $A \cup (B - A) = B$.
24. ★ The *product set* of sets A and B is the set of all ordered pairs (a, b) where $a \in A$ and $b \in B$; it is denoted by $A \times B$. Prove that $A \times (B \cap C) = (A \times B) \cap (A \times C)$.